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SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
Instructor(s) Name		Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju	
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week8 – Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 16.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question		Expected Time to complete

1	Lab 16 – Database Design and Queries: Schema Design and SQL Generation Lab Objectives: • Understand schema design principles for relational databases. • Use AI-assisted tools to generate SQL schema definitions. • Write and refine SQL queries (CRUD + aggregate operations). • Explore how AI can optimize queries and ensure normalization. Task 1 – Hospital Management Database Task: Create schema and queries for a Hospital Management System. Instructions: • Tables: Doctors, Patients, Appointments. • Use AI to define constraints (unique IDs, valid dates). • Generate queries: • List all appointments for a specific doctor. • Retrieve patient history by patient ID. • Count total patients treated by each doctor. Expected Output: • Normalized schema and SQL queries with joins. CODE: DROP TABLE IF EXISTS Appointments; DROP TABLE IF EXISTS Patients; DROP TABLE IF EXISTS Doctors; CREATE TABLE Doctors (doctor_id INT PRIMARY KEY, name VARCHAR(100), specialization VARCHAR(100)); CREATE TABLE Patients (patient_id INT PRIMARY KEY, name VARCHAR(100), age INT, gender VARCHAR(10)); CREATE TABLE Appointments (appointment_id INT PRIMARY KEY, doctor_id INT, patient_id INT, appointment_date DATE, FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id), FOREIGN KEY (patient_id) REFERENCES Patients(patient_id)); INSERT INTO Doctors VALUES	Week8 Wednesday
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	(1, 'Dr. Smith', 'Cardiology'),	
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	<pre>(2, 'Dr. John', 'Neurology'); INSERT INTO Patients VALUES (101, 'Alice', 25, 'Female'), (102, 'Bob', 30, 'Male'); INSERT INTO Appointments VALUES (1, 1, 101, '2025-08-01'), (2, 1, 102, '2025-08-02'), (3, 2, 101, '2025-08-03'); SELECT * FROM Doctors; SELECT * FROM Patients; SELECT * FROM Appointments; SELECT a.appointment_id, a.appointment_date, p.name AS patient_name FROM Appointments a JOIN Patients p ON a.patient_id = p.patient_id WHERE a.doctor_id = 1; SELECT a.appointment_id, a.appointment_date, d.name AS</pre>	
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```
doctor_name
FROM Appointments a
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE a.patient_id = 101;

SELECT d.name, COUNT(DISTINCT a.patient_id) AS total_patients
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.name;
```

Task 2 – Real-Time Application: Online Shopping Database Scenario:
Design a database for an E-commerce platform.

Requirements:

- Tables: Users, Products, Orders, OrderDetails.
- Generate queries to:
- Retrieve all orders by a user.
- Find the most purchased product.
- Calculate total revenue in a given month.
- AI should also suggest normalization improvements and query optimization.

Expected Output:

- Complete schema with relationships + SQL queries for analytics.

CODE:

```
DROP TABLE IF EXISTS OrderDetails;
```

```

DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
  user_id INT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100) UNIQUE
);

CREATE TABLE Products (
  product_id INT PRIMARY KEY,
  name VARCHAR(100),
  price DECIMAL(10,2)
);

CREATE TABLE Orders (
  order_id INT PRIMARY KEY,
  user_id INT, order_date DATE,
  FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE OrderDetails (
  order_detail_id INT PRIMARY KEY,
  order_id INT, product_id INT,
  quantity INT,
  FOREIGN KEY (order_id) REFERENCES Orders(order_id),
  FOREIGN KEY (product_id) REFERENCES Products(product_id)
);

INSERT INTO Users VALUES
(1, 'Alice', 'alice@email.com'),
(2, 'Bob', 'bob@email.com');

INSERT INTO Products VALUES
(1, 'Laptop', 50000),
(2, 'Phone', 20000);

INSERT INTO Orders VALUES
(1, 1, '2025-08-01'),
(2, 1, '2025-08-10'),
(3, 2, '2025-08-15');

INSERT INTO OrderDetails VALUES
(1, 1, 1, 1),
(2, 2, 2, 2),
(3, 3, 2, 1);

```

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```

-- 1. Orders by user
SELECT * FROM Orders WHERE user_id = 1;

-- 2. Most purchased product
SELECT p.name, SUM(od.quantity) AS total_sold
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
GROUP BY p.name
ORDER BY total_sold DESC
LIMIT 1;

-- 3. Total revenue (August 2025)
SELECT SUM(p.price * od.quantity) AS total_revenue
FROM Orders o
JOIN OrderDetails od ON o.order_id = od.order_id
JOIN Products p ON od.product_id = p.product_id
WHERE MONTH(o.order_date) = 8 AND YEAR(o.order_date) =
2025;

```

Task 3 – Library Database Task:

Design schema for a Library Management System.

Instructions:

- Tables: Books, Members, Loans.
- Use AI to suggest indexing strategy for faster queries.
- Generate queries:
- Retrieve all books currently issued.
- Find overdue books (loan date > 30 days).
- Count number of books loaned by each member.

Expected Output:

- Schema + SQL queries demonstrating joins and conditions.

CODE:

```

DROP TABLE IF EXISTS Loans;
DROP TABLE IF EXISTS Members;
DROP TABLE IF EXISTS Books;

```

```

CREATE TABLE Books (
  book_id INT PRIMARY KEY,
  title VARCHAR(100),
  author VARCHAR(100)
);

```

	<pre>CREATE TABLE Members (member_id INT PRIMARY KEY, name VARCHAR(100));</pre>	
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```
CREATE TABLE Loans (
```


	<pre>loan_id INT PRIMARY KEY, book_id INT, member_id INT, loan_date DATE, return_date DATE, FOREIGN KEY (book_id) REFERENCES Books(book_id), FOREIGN KEY (member_id) REFERENCES Members(member_id)); INSERT INTO Books VALUES (1, 'DBMS', 'Author A'), (2, 'AI Basics', 'Author B'); INSERT INTO Members VALUES (1, 'John'), (2, 'Alice'); INSERT INTO Loans VALUES (1, 1, 1, '2025-07-01', NULL), (2, 2, 2, '2025-06-01', NULL); -- 1. Books currently issued SELECT b.title, m.name FROM Loans l JOIN Books b ON l.book_id = b.book_id JOIN Members m ON l.member_id = m.member_id WHERE l.return_date IS NULL; -- 2. Overdue books (>30 days) SELECT b.title, m.name FROM Loans l JOIN Books b ON l.book_id = b.book_id JOIN Members m ON l.member_id = m.member_id WHERE DATEDIFF(CURDATE(), l.loan_date) > 30 AND l.return_date IS NULL; -- 3. Books per member SELECT m.name, COUNT(l.book_id) AS total_books FROM Members m LEFT JOIN Loans l ON m.member_id = l.member_id GROUP BY m.name;</pre>	
	Task 4: Optimize Queries Instructions: • Use AI tools to analyze query efficiency.	

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| | <ul style="list-style-type: none">• Optimize inefficient queries using joins, indexes, or reducing subqueries.• Test optimized queries for correctness. | |
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Starter Code Example: -

- Original query

	<pre>SELECT * FROM Books WHERE author_id IN (SELECT author_id FROM Authors WHERE country='UK');</pre> -- Optimized query <pre>SELECT b.* FROM Books b JOIN Authors a ON b.author_id = a.author_id WHERE a.country = 'UK';</pre> Expected Output: • Both queries return the same result: all books written by UK authors. • Optimized query performs faster for larger datasets. CODE: <pre>SELECT * FROM Books WHERE author_id IN (SELECT author_id FROM Authors WHERE country='UK');</pre><pre>SELECT b.* FROM Books b JOIN Authors a ON b.author_id = a.author_id WHERE a.country = 'UK';</pre> Task 5: University Course Registration Scenario: SR University needs a course registration system for students enrolling in different courses under faculty supervision. • Use Copilot to design schema tables: Students, Courses, Faculty, and Registrations. • Define relationships: • Each student can register for multiple courses. • Each course is taught by a faculty member. • Write SQL queries to: • List all students enrolled in a specific course. • Find faculty members teaching more than 2 courses. • Retrieve courses with the highest number of registrations. CODE: <pre>DROP TABLE IF EXISTS Registrations; DROP TABLE IF EXISTS Courses; DROP TABLE IF EXISTS Faculty; DROP TABLE IF EXISTS Students;</pre><pre>CREATE TABLE Students (student_id INT PRIMARY KEY, name VARCHAR(100)</pre>	
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	);	
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	<pre>CREATE TABLE Faculty (faculty_id INT PRIMARY KEY, name VARCHAR(100)); CREATE TABLE Courses (course_id INT PRIMARY KEY, course_name VARCHAR(100), faculty_id INT, FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)); CREATE TABLE Registrations (registration_id INT PRIMARY KEY, student_id INT, course_id INT, FOREIGN KEY (student_id) REFERENCES</pre>	
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	<pre> Students(student_id), FOREIGN KEY (course_id) Courses(course_id)); INSERT INTO Students VALUES (1, 'Alice'), (2, 'Bob'), (3, 'Charlie'); INSERT INTO Faculty VALUES (1, 'Prof. A'), (2, 'Prof. B'); INSERT INTO Courses VALUES (101, 'DBMS', 1), (102, 'AI', 1), (103, 'Networks', 2); INSERT INTO Registrations VALUES (1, 1, 101), (2, 2, 101), (3, 3, 102), (4, 1, 102), (5, 2, 103); -- 1. Students in a course SELECT s.name FROM Registrations r JOIN Students s ON r.student_id = s.student_id WHERE r.course_id = 101; </pre>	REFERENCES
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	<pre>-- 2. Faculty teaching >2 courses SELECT f.name, COUNT(c.course_id) AS total_courses FROM Faculty f JOIN Courses c ON f.faculty_id = c.faculty_id GROUP BY f.name HAVING COUNT(c.course_id) > 2; -- 3. Course with highest registrations SELECT c.course_name, COUNT(r.student_id) total_students FROM Courses c JOIN Registrations r ON c.course_id = r.course_id GROUP BY c.course_name ORDER BY total_students DESC LIMIT 1;</pre>	AS
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